

User Study

Instructions for Grading. **PLEASE READ CAREFULLY.**

Overview

Imagine you are a user interacting with a chatbot service (e.g., Gemini, ChatGPT, or DeepSeek) for help with programming tasks. On each page, you will see:

- **A Question:** This could be a vague user query or a code snippet requiring clarification or completion. Questions containing code will be displayed in code blocks for clarity.
- **Two Responses:** labeled Answer A and Answer B

Your Task

Please evaluate Answer A and Answer B based on the following criteria:

- **Precision and Focus:** How directly the response targets the question's core issue, using clear guidance, examples, or code.
- **Contextual Fit:** How well the response matches the user's programming goal (e.g., debugging, feature implementation).
- **Answer Faithfulness:** How effectively the response provides actionable steps or code to achieve the user's desired outcome.
- **Correctness:** Correctness in LLM responses refers to the accuracy and factual validity of the information, code, or guidance provided, ensuring it is free from errors and aligns with the user's query and intended outcome.

How to Rate

For each question, you will provide two types of ratings:

1. **Preference Rating (1–5 scale):**
 - **"1"**: Answer A is better, based on the criteria above.
 - **"5"**: Answer B is better, based on the criteria above.
 - **"3"**: Both answers are equally good, based on the criteria above.

Note:

- **Incomplete Responses:** Some responses may appear cut off due to a 256-word limit imposed on the chatbots. Please grade only the content provided, ignoring any abrupt endings or non-English text (e.g., stray characters).
- **Code Formatting:** Questions and responses containing code will be presented in code blocks or artifact tags for clarity. Ensure you review these carefully for correctness and relevance.
- **Consistency:** Try to apply the same standards across all evaluations to ensure fair comparisons.

Question 1

JavaScript

```
const data = [1, 2, 3];
const result = data.map(num => num * 2);
console.log(result); // TODO: handle empty array
```

Answer A:

JavaScript

```
const data = [1, 2, 3];
const result = data.map(num => num * 2);
console.log(result);
```

Explanation:

The code snippet you provided is already correct and handles the case of an empty array. Let's break it down:

1. `const data = [1, 2, 3];` declares an array with numbers 1, 2, and 3.
2. `data.map(num => num * 2);` applies the function `num => num * 2` to each element, doubling each number, and returns a new array.
3. If `data` is empty, `map()` returns an empty array, so no special handling is needed for empty arrays.

Answer B:

"Extra context:

User Question:

JavaScript

```
const data = [1, 2, 3];
const result = data.map(num => num * 2);
console.log(result); // TODO: handle empty array
```

AI clarification question: What is the intended behavior when the input array `data` is empty? Are you expecting an error or a specific default behavior (e.g., return an empty array)?

User follow up response: I want it to return an empty array if the input is empty."

AI Final answer:

You are absolutely correct! The intended behavior when the input array `data` is empty is to return an empty array. The `map()` method naturally returns an empty array when called on an empty array, so no additional handling is necessary.

Here is the corrected and complete code:

JavaScript

```
const data = [1, 2, 3];
const result = data.map(num => num * 2);
console.log(result);
```

If `data` were empty (`const data = [];`), `result` would also be an empty array `[]`.
Excellent job!

How to rate:

"1": strongly prefer answer A

"5": strongly prefer answer B

"3": both answers are equally good

Precision and Focus (1-5):3

Contextual Fit (1-5):3

Answer Faithfulness (1-5):3

Correctness (1-5):3

Question 2

I have an issue with my API call.

Answer A:

"Extra context:

User Question: *I have an issue with my API call.*

AI clarification question: *What API are you using? What specific error message or problem are you encountering?*

User follow up response: *I'm using the Twitter API, and I'm getting a '401 Unauthorized' error."*

AI Final answer:

A 401 Unauthorized error with the Twitter API is a common issue. It means your request is not properly authenticated.

To troubleshoot:

1. Verify your API key/token is correct and has not expired.
2. Check that you are including the token properly in the request headers.
3. Confirm you have the required permissions for the endpoint.
4. Ensure your system clock is accurate (OAuth tokens can fail if clocks are out of sync).
5. Review Twitter's API documentation for authentication requirements.

If you provide your request code and headers (without sensitive info), I can help you debug further.

Answer B:

Okay, I'm here to help! Please tell me about the issue you're having with your API call. The more detail you can provide, the better I can understand the problem and offer helpful solutions. Let's break it down.

Here's a good way to start – please answer these questions:

1. What API are you using? (e.g., Google Maps API, Twitter API, Stripe API, a custom API, etc.)
2. What are you trying to do? (Be specific! What endpoint are you calling? What data are you expecting to receive?)
3. What is the exact error message you're seeing? (Please copy and paste the entire error message.)
4. What have you already tried to fix the issue? (Checked docs, parameters, rate limits?)
5. Can you provide a sample of the data you're sending?
6. What is your environment? (e.g., browser, Node.js, etc.)

How to rate:

"1": strongly prefer answer A

"5": strongly prefer answer B

"3": both answers are equally good

Precision and Focus (1-5): 2

Contextual Fit (1-5): 1

Answer Faithfulness (1-5): 1

Correctness (1-5):3

Question 3

I'm confused about a concept in programming.

Answer A:

"Extra context:

User Question: *I'm confused about a concept in programming.*

AI clarification question: *Could you specify which programming concept is confusing you? For example, is it recursion, object-oriented programming, or something else?*

User follow up response: *I'm confused about recursion in Python, especially how the call stack works."*

AI Final answer:

Let's clarify recursion in Python.

What is recursion?

It's when a function calls itself to solve smaller instances of the same problem.

How does the call stack work?

Each recursive call is placed on the call stack, which keeps track of active function calls. When a function returns, it is popped off the stack.

Understanding the call stack helps you see why base cases are essential to prevent infinite recursion.

If you'd like, I can provide example code and diagrams to illustrate this further.

Answer B:

Okay, let's tackle this! I understand that feeling of being confused – it's incredibly common. To help me understand *exactly* what you're struggling with, I need a little more information.

Let's start with a few questions:

1. What specific concept are you confused about?
2. What programming language are you using?
3. What have you tried so far?
4. Can you give a specific example where you get stuck?

Don't worry if you don't have all the answers! We'll work through it together.

How to rate:

"1": strongly prefer answer A

"5": strongly prefer answer B

"3": both answers are equally good

Precision and Focus (1-5):2

Contextual Fit (1-5):2

Answer Faithfulness (1-5):1

Correctness (1-5):3

Question 4

Java

```
public class Example {  
    private int value;  
  
    public Example(int value) {  
        self.value = value;  
    }  
}  
  
// TODO: Add getters
```

Answer A:

Java

```
public class Example {  
    private int value;  
  
    public Example(int value) {  
        self.value = value;  
    }  
  
    // Getters  
    public int getValue() {  
        return self.value;  
    }  
  
    public void setValue(int value) {  
        self.value = value;  
    }  
}
```

Answer B:

"Extra context:

User Question:

```
Java
public class Example {
    private int value;

    public Example(int value) {
        self.value = value;
    }
}

// TODO: Add getters
```

AI clarification question: What is the intended use of the `value` field? What functionality do you want it to represent?

User follow up response: I want to execute a function `run()` – empty body for now; when `value` is equal to 57."

AI Final answer:

The following Java class `Example` encapsulates a private integer field `value` and provides controlled access and behavior based on its value.

- The constructor initializes the `value` field.
- The `getValue()` method is a standard getter that returns the current value.
- The `run()` method is defined with an empty body for now, serving as a placeholder for future functionality.
- The `checkValue()` method checks if `value` equals 57, and if so, invokes the `run()` method.

This design allows you to encapsulate the logic that should be triggered when `value` reaches a specific condition, keeping the class modular and easy to extend.

Java

```
public class Example {  
    private int value;  
    public Example(int value) {  
        this.value = value;  
    }  
    public int getValue() {  
        return value;  
    }  
    public void run() {  
        // empty for now  
    }  
    public void checkValue() {  
        if (value == 57) {  
            run();  
        }  
    }  
}
```

You can later implement the desired behavior inside the `run()` method as your application requirements evolve.

How to rate:

"1": strongly prefer answer A

"5": strongly prefer answer B

"3": both answers are equally good

Precision and Focus (1-5):5

Contextual Fit (1-5): 4

Answer Faithfulness (1-5):5

Correctness (1-5):3

Question 5

JavaScript

```
const numbers = [1, 2, 3, 4];  
const doubled = numbers.map(num => num * 2);  
console.log(doub); // Typo in variable name
```

Answer A:

Extra context:

User Question:

JavaScript

```
const numbers = [1, 2, 3, 4];  
const doubled = numbers.map(num => num * 2);  
console.log(doub); // Typo in variable name
```

AI clarification question: What output do you expect from this code?

User follow up response: I expect to print `` but there's a typo in the variable name.

AI Final answer:

You are correct! The variable `doub` is a typo. The correct variable name is `doubled`. Here's the fixed code:

JavaScript

```
const numbers = [1, 2, 3, 4];  
const doubled = numbers.map(num => num * 2);  
console.log(doubled); // Prints: [2, 4, 6, 8]
```

This will output the expected doubled array.

Answer B:

You're absolutely right! The typo in the variable name `doub` causes the problem. The code intends to print the doubled array stored in `doubled`.

Corrected code:

JavaScript

```
const numbers = [1, 2, 3, 4];  
const doubled = numbers.map(num => num * 2);  
console.log(doubled);
```

How to rate:

"1": strongly prefer answer A

"5": strongly prefer answer B

"3": both answers are equally good

Precision and Focus (1-5):3

Contextual Fit (1-5):3

Answer Faithfulness (1-5):3

Correctness (1-5):3